Detection of Suspicious Activities in Public Spaces Using Video Surveillance

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Code: https://github.com/mat-tp/suspicious-activity-detection

May 13, 2026



Outline I

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Motivation

- Surveillance cameras are everywhere – train stations, malls, streets.
- Human operators cannot watch all feeds simultaneously; fatigue reduces attention.
- Need automatic activity recognition to assist operators.
- Deep learning achieves high accuracy but requires:
  - GPU hardware
  - Huge labelled datasets (e.g., ActivityNet, YouTube-8M)
  - Black-box decisions – difficult to interpret in security contexts.

Goal: A classical computer vision pipeline that is lightweight, interpretable, and robust to poor video quality.



Contributions

1. Activity classification under authentic distortions (focus, exposure, focus+exposure).
2. First repurposing of the AD-SVD dataset *for classification* (originally designed for tracking).
3. Head-to-head comparison of a from-scratch Random Forest with scikit-learn's implementation.
4. Dual-mode evaluation: custom MOG2 detector vs. dataset's ground-truth bounding boxes to measure the impact of detection robustness.



AD-SVD Dataset Overview

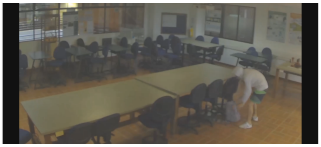
- **4,476 videos**, 8 suspicious activities, 4 distortion types.
- Recorded indoors/outdoors, 7–10 FPS, fixed surveillance camera.
- **Activities:** Fighting in Group, Leaving Package, Passing Out, Person Pushing, Person Running, Prowl, Robbery with Knife, Walking.
- **Distortions:** Pristine (3.6%), Exposure (32.3%), Focus (31.5%), Focus+Exposure (32.3%).
- Ground-truth bounding box for a single primary actor per frame.
- Publicly available: IEEE Dataport.

(Gomez-Nieto et al., 2022; AD-SVD)

Dataset Preview – Activity Classes



Fighting in Group



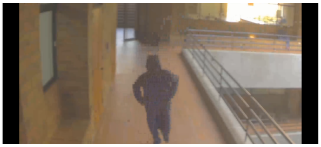
Leaving Package



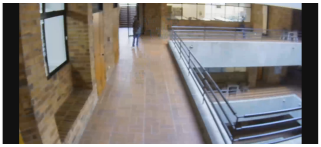
Passing Out



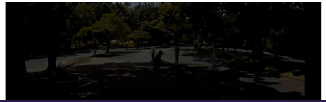
Person Pushing

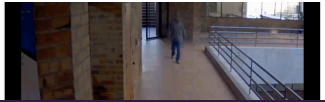


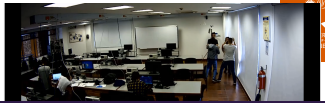
Person Running



Prowl



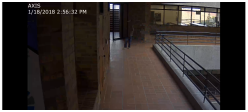




Dataset Preview – Distortion Types



Pristine



Exposure



Focus



Focus+Exposure

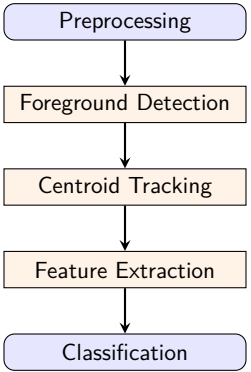
- **Focus blur** diffuses object boundaries → detection often fails.
- **Exposure changes** reduce foreground-background contrast.
- Almost **96%** of videos are distorted — testing ground for robustness.

Challenge: Distortion Effects

Distortion Type	Videos	% of total
Focus and Exposure (ExFo)	1,447	32.3%
Exposure (Exp)	1,445	32.3%
Focus (Fo)	1,409	31.5%
Pristine (Pri)	160	3.6%



Pipeline Overview



Stages: Pre-processing → Foreground Detection → Centroid Tracking → Optical Flow → Feature Extraction → Classifier.

Detection Modes

MOG2 (our detector)

- Gaussian Mixture Model background subtraction.
- Morphological cleaning, blob size filtering (> 500 px^2).
- Can produce *multiple* detections per frame.
- Fails on most Focus & ExFo videos (blur hides foreground).

MOG2 yields only 848 usable videos (19%)

Ground-truth (GT) boxes

- One bounding box per frame, from human annotation.
- Exactly *one* track per video.
- Unaffected by distortion.
- Used to study the upper bound of classification performance.

GT yields 4,461 videos (99.7%).

Tracking & Feature Extraction

Centroid Tracking

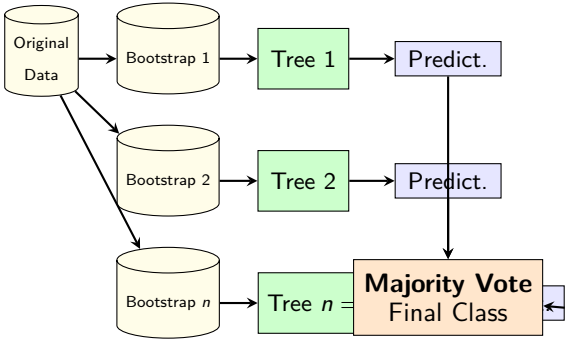
- Greedy nearest-neighbour match using Euclidean distance.
- Tracks shorter than 30 frames are discarded → removes spurious background blobs.

14 Per-track Features

- Trajectory geometry: length, total distance, net displacement, straightness, area.
- Speed statistics: mean, max, std, skewness, kurtosis.
- Direction statistics: mean and std of heading angle.
- Global optical flow: mean magnitude & angle (Lucas-Kanade).

Per-video aggregation: mean, max, std of each per-track feature + number of tracks → 43 features per video.

Random Forest: A Quick Overview



How it works

- Ensemble of many decision trees.
- Each tree trained on a bootstrap sample

Why we use it

- Resists overfitting, handles small sample sizes.

Classifiers

Four models evaluated on the same 43-dimensional features:

- **SVM** (RBF kernel, $C = 10$, balanced class weights).
- **K-Nearest Neighbours** ($k = 10$).
- **Random Forest** – scikit-learn (200 trees, balanced weights, unconstrained depth).
- **Custom Random Forest** – built from scratch (200 trees, max depth 10, entropy split, bootstrap, $\sqrt{d}$ features per split).

Data split: 70% train, 15% validation, 15% test (stratified).
Features standardised inside a pipeline.

Activity Classification – GT Mode (4,461 videos)

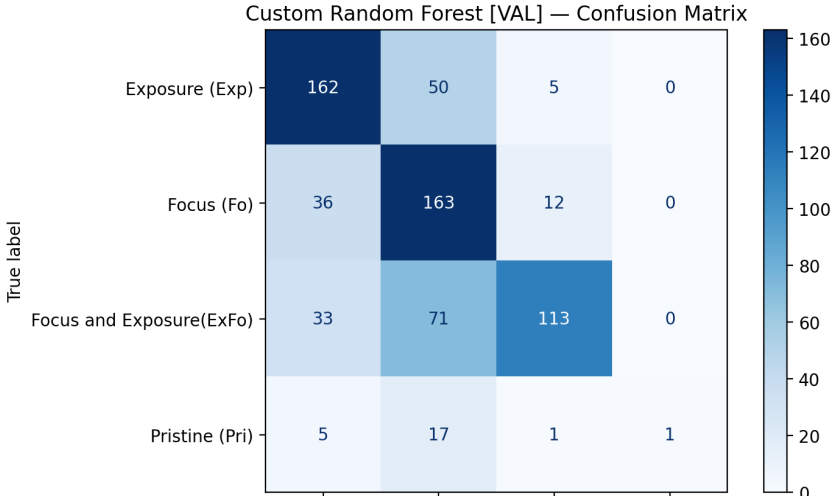
Classifier	Test Acc.	CV Mean
SVM (RBF)	76.4%	73.7%
KNN ($k = 10$)	69.6%	68.5%
RF (sklearn)	94.0%	91.1%
Custom RF	92.5% (val)	—
Random (8-cl.)	12.5%	—

Top per-class F1:

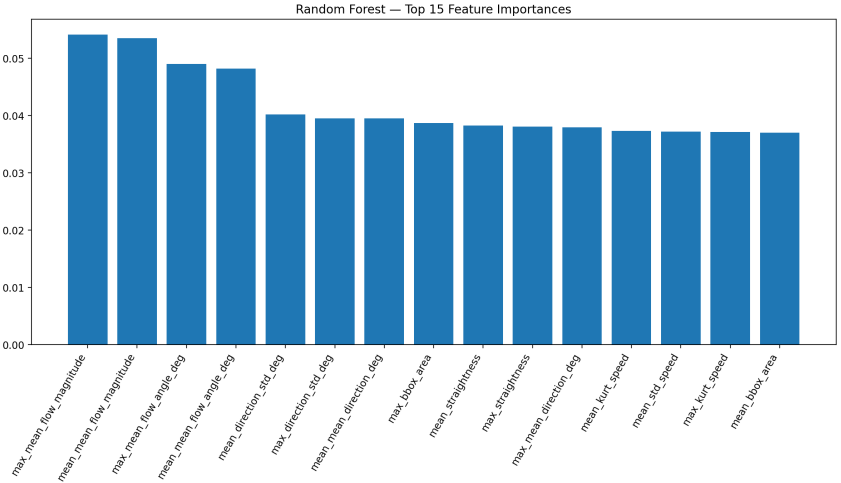
- Person Running: **0.99**
- Passing Out: 0.96
- Prowl: 0.96
- Fighting: 0.95
- Walking: 0.95

Macro avg F1 = 0.94

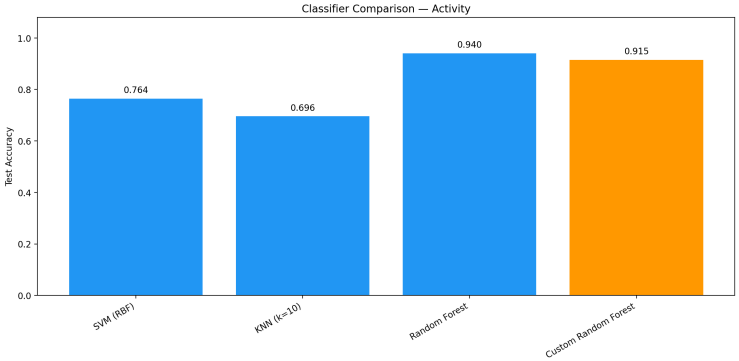
Confusion Matrix – GT Mode (sklearn RF)



Feature Importance (Random Forest)

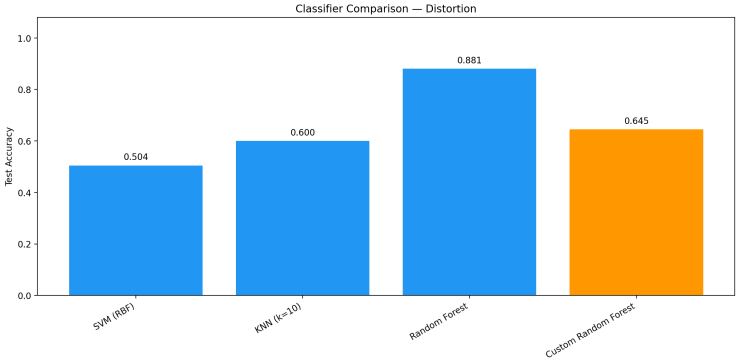


Accuracy Comparison – Activity Classification



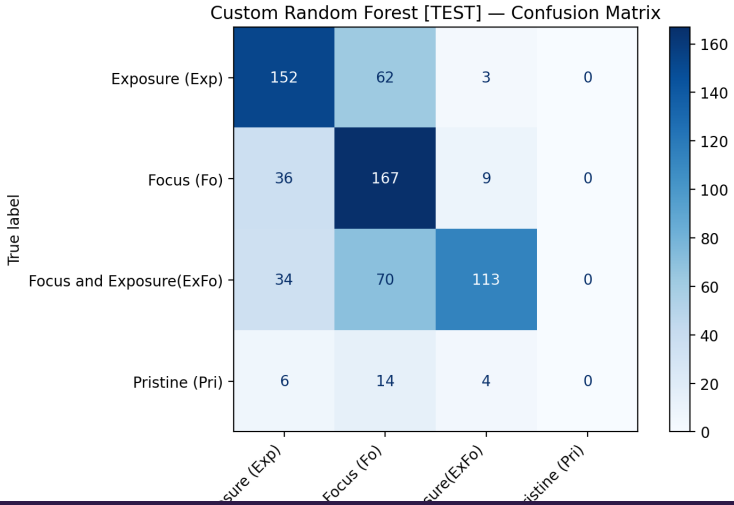
Comparison of all four classifiers on activity detection.

Accuracy Comparison – Distortion Classification



Comparison of all four classifiers on distortion type detection.

Custom Random Forest – Confusion Matrix



Activity Classification – MOG2 Mode (848 videos)

Severe class imbalance, only 128 test samples

Classifier	Test Acc.	CV Mean
SVM (RBF)	62.5%	60.2%
KNN ($k = 10$)	57.8%	55.9%
RF (sklearn)	71.9%	64.2%
Custom RF	68.4%	61.7%

- Despite losing 81% of videos to detection failures, motion features still carry discriminative signal.
- KNN suffers from 43-dimensional feature space on small training set.

Distortion Classification – GT Mode (4 classes)

Can we tell the distortion type from motion features alone?

Classifier	Test Acc.	CV Mean
SVM (RBF)	50.4%	46.7%
KNN ($k = 10$)	60.0%	55.9%
RF (sklearn)	88.1%	84.6%
Custom RF	64.5%	61.9%
Random (4-class)	25.0%	—

- Distortion alters track stability, speed variance, and number of detected tracks.
- **First report of distortion classification accuracy on AD-SVD.**

Detection – the Bottleneck

Property	MOG2 (custom)	GT boxes
Videos with tracks	848 (19%)	4,461 (99.7%)
Class balance	Severely imbalanced	Near-balanced
Best activity acc.	71.9%	94.0%

- 81% of videos filtered out primarily by focus blur.
- The 26-point accuracy gap is due to **detection failures**, not feature design.
- GT mode shows what the pipeline can achieve with **good detections**.



Key Findings

Beside the state of the art in activity classification :

1. Trajectory features (speed, straightness, spatial range) + simple Random Forest $\rightarrow$ 94% activity accuracy under clean detections.
2. Same features can classify the distortion type with 88.1% (4-class), without pixel-level quality inputs.
3. The from-scratch Random Forest is only 1.5–3.5 pp behind the library version, quantifying the cost of hand-rolling.
4. Detection robustness is the main bottleneck: MOG2 drops to 19% of videos.



Comparison with Deep Learning State-of-the-Art

Large-scale action recognition benchmarks (Kinetics-400, UCF101, HMDB51)

- **I3D** (Carreira & Zisserman, 2017) – inflated 3D ConvNets, top-1 $\sim$ 95% (Kinetics). Needs massive labelled videos and GPU.
- **SlowFast** (Feichtenhofer et al., 2019) – two-stream network for fine temporal resolution, state-of-the-art accuracy but high compute.
- **VideoMAE** (Tong et al., 2022) – masked auto-encoder with Vision Transformers, effective with limited data, but still GPU-heavy.
- **Video Vision Transformer (ViViT)** (Arnab et al., 2021) – pure transformer model, strong but data-hungry.

Our classical pipeline vs. deep learning

Our Random Forest + hand-crafted features

Typical deep learning approach

Limitations & Future Work

Limitations

- MOG2 sensitive to lighting, shadows, and focus blur.
- Simple centroid tracker cannot handle occlusions or identity switches.
- Features omit intra-track dynamics (acceleration, periodicity).

Future Work

- Add acceleration statistics, FFT periodicity, and circular direction variance.
- Kalman-filter or IoU-based tracker for occlusion resilience.
- Distortion-aware preprocessing (adaptive thresholding, deblurring).
- Real-time stream classification with existing GUI.



Thank You!

Questions?

